EVA: Aligning Video World Models with Executable Robot Actions via Inverse Dynamics Rewards

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Research Questions

- How can video world models for robotics be aligned to generate physically executable trajectories that respect kinematic constraints?
- Can an inverse dynamics model (IDM) be repurposed as a reward model to penalize executability violations during video generation training?
- What is the impact of IDM-based reward alignment on downstream robotic task execution success rates?

Core Idea

The paper addresses the "executability gap" in video world models for robotics: while generated videos may look visually plausible, they often contain kinematic violations (arm deformations, impossible joint movements) that produce unstable robot actions when decoded by inverse dynamics models. The key insight is to train an IDM on real robot data and repurpose it as a reward model that evaluates generated videos through their implied action sequences, penalizing non-smooth motions and embodiment constraint violations. This creates a dense training signal that aligns video generation with physical feasibility without requiring expensive rejection sampling at inference time.

Key Formula

IDM-based Penalty Function

$$\mathcal{P}(A) = \lambda_j \mathcal{P}_j + \lambda_\alpha \mathcal{P}_\alpha + \lambda_{v\text{-lim}} \mathcal{P}_{\text{vel}} + \lambda_{a\text{-lim}} \mathcal{P}_{\text{acc}}$$

Total penalty P(A) combines jerk penalty P_j, acceleration penalty P_α, velocity limit violations P_vel, and acceleration limit violations P_acc, each weighted by lambda coefficients. This penalty is computed from IDM-predicted action sequences and captures kinematic implausibility.

Executability Reward

$$R(V) = \left(1 + \frac{\mathcal{P}(A)}{P_0}\right)^{-\gamma}$$

Reward R(V) for video V is computed by mapping penalty P(A) into bounded reward using scale P_0 and decay rate γ. Higher penalties (more kinematic violations) result in lower rewards, providing training signal for GRPO alignment.

Key Figure

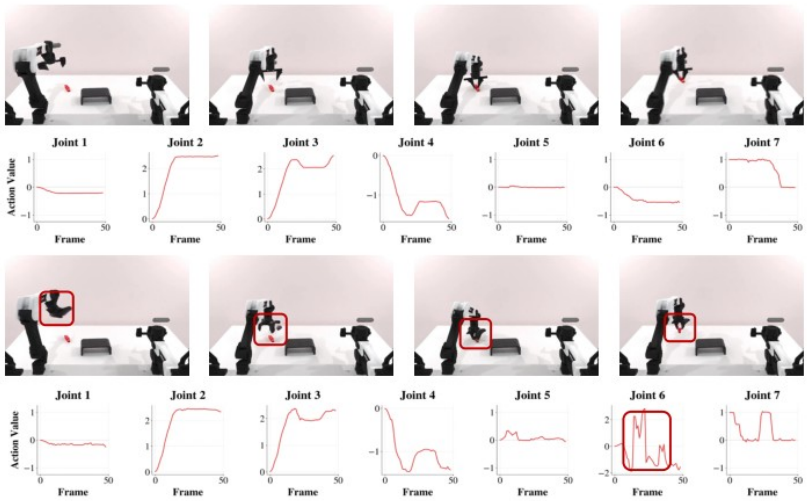

Fig. 2: Illustration of how visual artifacts translate into kinematic violations. The plots display the 7-DOF joint angles (in radians) for the left arm, ordered from the base (Joint 1) to the gripper (Joint 7). **(Top)** A high-quality generation video. The translated actions are smooth and physically executable, yielding a high reward score of 7.94. **(Bottom)** A failure case exhibiting severe visual artifacts (highlighted in red). Consequently, the IDM translates these visual artifacts into erratic, high-frequency jitter, particularly visible in the distal joints (e.g., Joints 6 and 7), leading to a low reward of 3.04.

5.1 Experimental Settings

Base model. Our video world model is a latent video diffusion model based on the Diffusion Transformer (DiT) [34]. We instantiate it with the Wan2.1-14B backbone [42] and incorporate diffusion forcing [9] to improve rollout generation conditioned on observation history. We initialize from the Large Video Planner (LVP) checkpoint [10], which is pretrained on large-scale manipulation data, and then perform supervised fine-tuning (SFT) on our embodiment-specific dataset, yielding **Ours (w/o RL)**. We then apply GRPO post-training with the IDM-based executability reward in Section 4.2 to obtain **Ours**. The IDM is trained as in Section 4.1 and kept frozen during GRPO fine-tuning.

Baselines. For rollout-quality evaluation, we compare against Vidar [17], initialized from the Wan2.2-5B checkpoint [42] and fine-tuned under the same protocol and data. For simulation policy execution, we compare against strong imitation-learning and VLA baselines: ACT [48], Diffusion Policy (DP) [12], RDT [29], and π_0 [4]. For real-robot evaluation, we additionally include GE-Act [26].

Figure 2: Shows how visual artifacts in generated videos directly translate into erratic, high-frequency joint trajectories when decoded by the IDM, demonstrating the executability gap with concrete joint angle plots.

Limitations

- Reward focuses only on kinematic feasibility and smoothness, not contact dynamics like forces, friction, or torques critical for precision manipulation

- Diffusion-based video generation is computationally expensive, limiting applicability to high-frequency reactive control
- May exhibit reward hacking behaviors during prolonged GRPO training, requiring careful checkpoint selection and early stopping
- IDM spatial softmax architecture is crucial but may not generalize to all robot morphologies or action spaces